# Naive Bayes: Bernoulli and Multinomial MLE

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**Abstract**

This short note shows that the maximum-likelihood estimators for Bernoulli and Multinomial Naive Bayes admit closed-form solutions in both cases.

## 1 Introduction

Let

$$\mathcal{D} = \{(x^{(r)}, y_r)\}_{r=1}^{n}$$

be a labeled training set, and let

$$V = \{w_1, \ldots, w_d\}$$

be the vocabulary, e.g. the set of all words in the training set. Finally consider,

$$\mathcal{Y} = \{y_1, \ldots y_K\}$$

to be the set of all possible classes/labels.

For a fixed class $k$, define

$$A_k := \{r : y_r = k\}, \qquad N_k := |A_k|.$$

The Bernoulli and Multinomial versions of Naive Bayes correspond to two different probabilistic descriptions of a document.

The aim below is to derive the maximum-likelihood estimators of these two class-conditional probabilities, which serves as our machine learning *objective*. Interestingly, in both cases the optimization admits a *closed-form solution*, so fitting the model reduces to computing suitable empirical counts from the training set.

In *Bernoulli NB*, each coordinate records only whether a vocabulary word appears

$$X_j^{(r)} \in \{0, 1\}.$$

Thus the natural class-conditional quantity to estimate is

$$X_j \mid Y = k \sim \text{Bernoulli}(\theta_{k,j}), \qquad \theta_{k,j} = P(X_j = 1 \mid Y = k),$$

namely the probability that a class-$k$ document contains the word $w_j$.

In *Multinomial NB*, instead,

$$X_j^{(r)} \in \mathbb{N}_0$$

records the number of occurrences of $w_j$. Conditional on the class, the tokens of a document are modeled as i.i.d. categorical draws from a class-specific distribution over the vocabulary. The corresponding quantity

$$W \mid Y = k \sim \text{Categorical}(\theta_{k,1}, \ldots, \theta_{k,d}), \qquad \theta_{k,j} = P(W = w_j \mid Y = k).$$

Then the count vector

$$X^{(r)} \mid Y = k \sim \text{Multinomial}(m_r; \theta_{k,1}, \ldots, \theta_{k,d}).$$

We stress that, despite using the same notation, the parameter $\theta_{k,j}$ has a different probabilistic meaning in the two models; in Bernoulli NB it is a document-level occurrence probability, whereas in Multinomial NB it is a token-level probability.

At the dataset level, the labeled documents are assumed to be *i.i.d.* In the Bernoulli model, the word-presence indicators are additionally assumed *conditionally independent given the class.* In the Multinomial model, the conditional independence assumption is instead imposed at the token level; the resulting word-count vector has a multinomial distribution.

Under these assumptions, the maximum-likelihood problem can be treated separately for each class $k$. Thus, for both models, it is enough to fix a class $k$ and optimize the corresponding parameter vector $\theta_k = (\theta_{k,1}, \ldots, \theta_{k,d})$. For the Bernoulli NB in particular, the conditional independence allows us to estimate each $\theta_{k,j}$ separately too. See Section A for more details.

## 2 Bernoulli Naive Bayes

For Bernoulli NB,

$$x_j^{(r)} = \begin{cases} 1, & \text{if word } w_j \text{ occurs in document } r, \\ 0, & \text{otherwise.} \end{cases}$$

For each class $k$ and word $w_j$, let

$$\theta_{k,j} = P(X_j = 1 \mid Y = k).$$

Thus

$$X_j \mid Y = k \sim \text{Bernoulli}(\theta_{k,j}).$$

For fixed $k, j$, the likelihood is

$$L(\theta_{k,j}) = \prod_{r \in A_k} \theta_{k,j}^{x_j^{(r)}} (1 - \theta_{k,j})^{1 - x_j^{(r)}}.$$

Hence

$$\ell(\theta_{k,j}) = \sum_{r \in A_k} \left[ x_j^{(r)} \log \theta_{k,j} + (1 - x_j^{(r)}) \log(1 - \theta_{k,j}) \right].$$

Differentiating and setting the derivative equal to zero gives

$$\frac{\sum_{r \in A_k} x_j^{(r)}}{\theta_{k,j}} - \frac{N_k - \sum_{r \in A_k} x_j^{(r)}}{1 - \theta_{k,j}} = 0,$$

where $N_k; = |A_k|$ denotes the number of all training documents belonging to class $k$.

Therefore,

$$\boxed{\theta_{k,j}^{*,B} = \frac{\sum_{r \in A_k} x_j^{(r)}}{N_k}.}$$

Thus

$$\boxed{\theta_{k,j}^{*,B} = \frac{\#\{\text{class-}k \text{ documents containing } w_j\}}{\#\{\text{class-}k \text{ documents}\}}.}$$

## 3 Multinomial Naive Bayes

For Multinomial NB,

$$x_j^{(r)} = \text{number of occurrences of } w_j \text{ in document } r.$$

For each class $k$, let

$$\theta_k = (\theta_{k,1}, \ldots, \theta_{k,d}), \qquad \sum_{j=1}^{d} \theta_{k,j} = 1,$$

where

$$\theta_{k,j} = P(W = w_j \mid Y = k).$$

If document $r$ has

$$m_r = \sum_{j=1}^{d} x_j^{(r)}$$

tokens, then

$$X^{(r)} \mid Y = k \sim \text{Multinomial}(m_r; \theta_{k,1}, \ldots, \theta_{k,d}).$$

Hence, up to factors independent of $\theta_k$,

$$L(\theta_k) \propto \prod_{r \in A_k} \prod_{j=1}^{d} \theta_{k,j}^{x_j^{(r)}}.$$

Therefore,

$$\ell(\theta_k) = \sum_{j=1}^{d} \left( \sum_{r \in A_k} x_j^{(r)} \right) \log \theta_{k,j} + C.$$

We maximize this subject to

$$\sum_{j=1}^{d} \theta_{k,j} = 1.$$

Using a Lagrange multiplier,

$$\mathcal{L} = \sum_{j=1}^{d} \left( \sum_{r \in A_k} x_j^{(r)} \right) \log \theta_{k,j} - \lambda \left( \sum_{j=1}^{d} \theta_{k,j} - 1 \right).$$

The first-order conditions give

$$\theta_{k,j} = \frac{\sum_{r \in A_k} x_j^{(r)}}{\lambda}.$$

The condition

$$\frac{\partial \mathcal{L}}{\partial \lambda} = 0$$

gives

$$\sum_{j=1}^{d} \theta_{k,j} = 1,$$

d therefore

$$\lambda = \sum_{r \in A_k} \sum_{\ell=1}^{d} x_\ell^{(r)}.$$

Hence

$$\boxed{\theta_{k,j}^{*,M} = \frac{\sum_{r \in A_k} x_j^{(r)}}{\sum_{r \in A_k} \sum_{\ell=1}^{d} x_\ell^{(r)}}.}$$

Thus

$$\boxed{\theta_{k,j}^{*,M} = \frac{\#\{\text{occurrences of } w_j \text{ in class } k\}}{\#\{\text{all token occurrences in class } k\}}.}$$

# A Independence and MLE

These assumptions also make the maximum-likelihood problem separable across classes. If $\theta = (\theta_1, \ldots, \theta_K)$, with $\theta_k$ denoting the parameters of the class-$k$ conditional model, then the class-conditional part of the log-likelihood can be written as

$$\ell(\theta; \mathcal{D}) = \sum_{k=1}^{K} \sum_{r \in A_k} \log p(x^{(r)} \mid Y = k; \theta_k).$$

This is true from the

Since the term corresponding to class $k$ depends only on $\theta_k$

$$\boxed{\arg\max_{\theta} \ell(\theta; \mathcal{D}) = \left( \arg\max_{\theta_1} \ell_1(\theta_1), \ldots, \arg\max_{\theta_K} \ell_K(\theta_K) \right),}$$

where

$$\ell_k(\theta_k) = \sum_{r \in A_k} \log p(x^{(r)} \mid Y = k; \theta_k).$$

Thus the likelihood parameters can be estimated *independently* for each class.

For Bernoulli NB, *conditional independence* gives the further factorization

$$p(x^{(r)} \mid Y = c_k; \theta_k) = \prod_{j=1}^{d} p(x_j^{(r)} \mid Y = c_k; \theta_{k,j}),$$

so the optimization also separates across vocabulary words. Hence each $\theta_{k,j}$ can be estimated independently.

For Multinomial NB, the word counts are *not* independent, since

$$\sum_{j=1}^{d} x_j^{(r)} = m_r.$$

However, the multinomial log-likelihood has the form

$$\ell_k(\theta_k) = C + \sum_{j=1}^{d} \left( \sum_{r \in A_k} x_j^{(r)} \right) \log \theta_{k,j},$$

subject to

$$\sum_{j=1}^{d} \theta_{k,j} = 1.$$

Thus we again obtain one parameter $\theta_{k,j}$ for each class and each word, but the parameters within a class are optimized jointly under the normalization constraint.

We therefore can focus on deriving the closed-form estimators

$$\theta_{k,j}^{*}.$$